

Negation-Aware Neural Information Retrieval

Do Dense Retrievers Understand “without”, “not”, and “less than” in Queries?

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Project Description

This project evaluates whether neural IR models handle negation in user queries. Using the NevIR benchmark (Weller et al., EACL 2024), we test BM25, TF-IDF, and Sentence-BERT on passage pairs differing only by a negation term. Each query requires distinguishing, for example, “drugs *without* aspirin” from “drugs *with* aspirin.” A negation-type classifier categorizes queries into explicit (“not”), implicit (“without”), and comparative (“less than”) types. The system is evaluated using Pairwise Accuracy and MRR, comparing all models against a 50% random baseline.

Problem and Motivation

Standard retrieval models rank documents by term relevance. If a user searches “*drugs without aspirin*,” BM25 and dense bi-encoders treat “aspirin” as a positive signal and rank aspirin-containing documents higher, the exact opposite of user intent. The NevIR paper showed that most state-of-the-art IR models perform at or below random chance on negation queries. By diagnosing which negation types fail and why, this project provides a reproducible benchmark motivating future negation-aware retrieval systems.

System Pipeline

1. **Load NevIR Dataset:** Use HuggingFace datasets library to load orionweller/NevIR. Each instance contains two queries differing by negation and two contrastive passage pairs.
2. **BM25 Sparse Retrieval:** Index passages using rank_bm25; rank each passage pair per query; record pairwise accuracy at rank 1.
3. **TF-IDF Baseline:** Build TF-IDF vectors via scikit-learn; rank passages by cosine similarity; evaluate pairwise accuracy.
4. **Dense Retrieval (Sentence-BERT):** Encode queries and passages using all-MiniLM-L6-v2; rank by cosine similarity; compute Pairwise Accuracy and MRR.
5. **Negation Classifier:** Train a logistic regression classifier on query text to predict negation type; report per-type accuracy across all retrieval models.
6. **Evaluation and Comparison:** Report Pairwise Accuracy, MRR, and per-negation-type breakdown comparing all models against random baseline (50%).

Expected Outcome

We expect all three retrieval models to perform near or below 50% pairwise accuracy on negation queries, consistent with the NevIR paper’s findings. Explicit negation (“not”) may be handled slightly better than implicit (“without”) since it appears directly in query text. A per-category breakdown will confirm that negation is a systematic blind spot in current neural IR models.

Grading Criteria

Criterion	How We Address It
Implementation as per paper	Core pipeline: NevIR dataset loading, BM25 sparse retrieval, TF-IDF baseline, Sentence-BERT dense retrieval, pairwise ranking, and negation-type classification.
Similarity of results	Results table comparing Pairwise Accuracy and MRR of BM25, TF-IDF, and Sentence-BERT; cross-encoders expected to outperform bi-encoders, matching the paper's finding.
Code documentation	Modular Python code with docstrings, inline comments, and a README with instructions to reproduce all experiments and evaluation results.
Proposal + Report	This proposal; final report covering methodology, results, per-negation-type analysis, and limitations of current neural IR models on negation.

Tools, Dataset & Division of Work

Libraries: datasets, rank_bm25, sentence-transformers, scikit-learn, numpy, pandas, matplotlib.

Dataset: NevIR via HuggingFace (orionweller/NevIR) containing query-passage negation pairs. No GPU required for BM25 and TF-IDF baselines.

Reference: Weller, O., Lawrie, D., & Van Durme, B. (2024). NevIR: Negation in Neural Information Retrieval. In *Proceedings of EACL 2024*, pages 2274–2287. doi:10.18653/v1/2024.eacl-long.139